AEROTHON 2026 - HAL x IITI

Physics-Informed Digital Twin for Real-Time Four-Stage Turbojet Health Monitoring

Team Name: SHUNYA

Participants: LAKSHYA JAIN, SIYA JULKA,

DIVYA JALAN, RUDRA MISHRA

PROBLEM UNDERSTANDING & MOTIVATION

UNMEASURED DEGRADATION

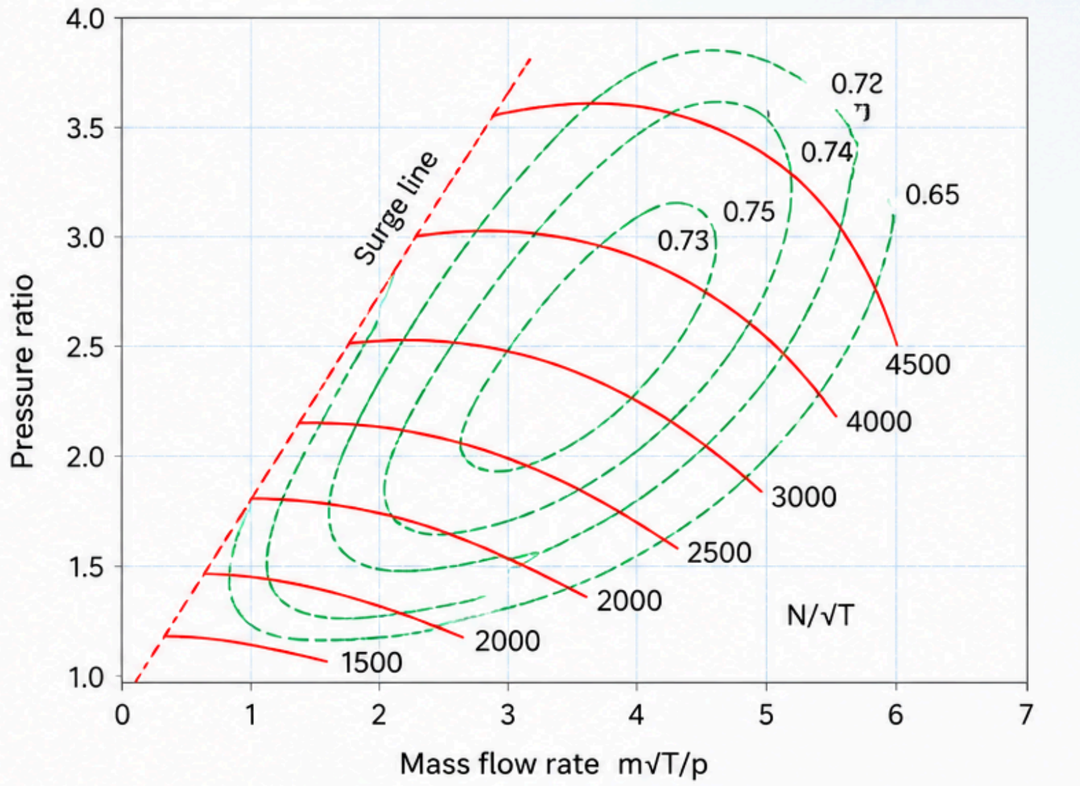
Component health indicators are not directly observable from onboard sensors during operation.

SIMULATION IS TOO SLOW

High-fidelity thermodynamic models are accurate but computationally unfit for real-time monitoring.

THE DIGITAL-TWIN OPPORTUNITY

Fusing sensor data with engineering knowledge enables continuous, interpretable state estimation.



Compressor degradation manifests as a progressive shift of the operating point on the compressor characteristic map, reducing pressure ratio, isentropic efficiency, and surge margin.
The proposed digital twin continuously estimates this shift from sparse sensor measurements.

LITERATURE REVIEW / EXISTING APPROACHES

WHAT OTHERS HAVE ALREADY TRIED

Older models (LSTM, CNN)
look at sensor history to guess how much life an engine has left.

A 2025 paper (Wu et al.)
built an improved Transformer, “TMR”, plus a 3D digital-twin dashboard, tested on NASA's CMAPSS dataset.

TMR beat older models on accuracy
but it’s a heavy, one-number “black box” health score.

Gap we found
no separate health score per part,
and no confidence/uncertainty on the answer.

PROPOSED METHODOLOGY & TECHNICAL APPROACH

1 Our Simple 3-Part Plan

Step 1 — Physics Math
Simple engine formulas check each part's health.

Step 2 — ML Model
Random Forest predicts future readings, thrust & fuel efficiency.

Step 3 — Confidence
Run model multiple times and measure answer spread.

Everything stays explainable — no hidden black box.

1. Physics Formulas → 2. ML Model Random Forest → 3. Confidence (Uncertainty)

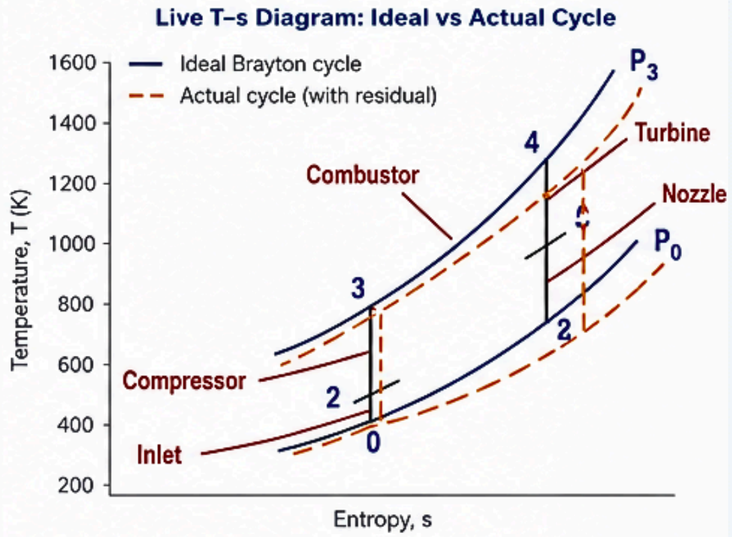
2 Thermodynamic Analysis: Brayton Cycle Validation

SENSOR (DATA) → PHYSICS SOLVER → RESIDUAL LEARNING → PLOT T–s CYCLE

Live sensor pairs (P0/T0 → P2/T2 → P3/T3 → P4/T4) feed a physics solver.

Solver reconstructs actual T–s cycle and ideal cycle.

Efficiency gap (residual) is tracked and drives health scores.



Orange gap = efficiency loss to real-world degradation — the same signal that drives the health-index dashboard.

DATA SOURCES & PREPROCESSING

03 — DATASET

Synthetic, physics-based data across four stages and multiple engines



- *Many virtual engines, each recorded over several flight cycles as they slowly degrade.*
- *Cleaning steps: check missing values, remove outliers, and normalize (Z-score) so every sensor is on the same scale.*
- *The first 10 cycles of each engine = the "healthy baseline" we compare everything else against.*

MODEL ARCHITECTURE / IMPLEMENTATION PLAN

SENSOR INGESTION
RPM, fuel flow, P2/P3, P4/T4, P4/P0, flight condition

STATE ESTIMATION
Physics model + learned residuals

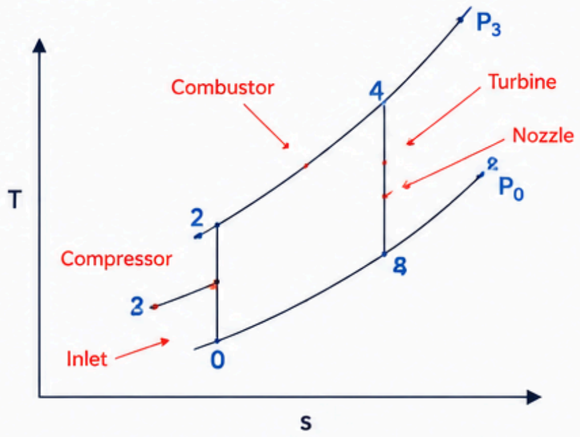
HEALTH INDICES
Compressor • Combustor • Turbine • Overall

PREDICTION LAYER
Thrust, efficiency, degradation trend & uncertainty

Continuously updated as new cycles arrive — the twin refines itself over the engine's operating life

THERMODYNAMIC ANALYSIS

T = Temperature
p = pressure
s = entropy



IDEAL BRAYTON CYCLE

SENSOR (DATA) — Real-time engine measurements

PHYSICS SOLVER — Computes expected states

RESIDUAL LEARNING — Learns & corrects model residuals

PLOTTING OF A THERMODYNAMIC CYCLE — Assess performance & efficiency

EXPECTED OUTCOMES & EVALUATION METRICS

PERFORMANCE OUTPUTS

ENGINE THRUST
From turbine work and exhaust conditions

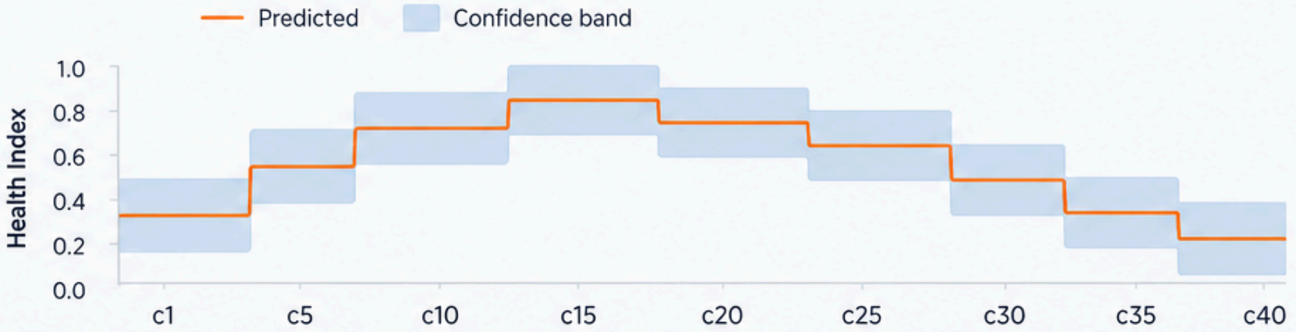
FUEL EFFICIENCY
TSFC trend vs. baseline, cycle over cycle

DEGRADATION TRAJECTORY
Projected health decline and remaining useful life



UNCERTAINTY QUANTIFICATION

An ensemble of surrogate models (or Monte-Carlo dropout) produces a spread of predictions per cycle. The spread is reported as a confidence band around every health index and performance estimate.



Health index over cycles — shaded band = confidence interval

TEAM COMPOSITION & CONTRIBUTIONS

Meet The Team

- *[Rudra Mishra] — Data collection & physics formulas (compressor / combustor / turbine health).*
- *[Lakshya jain] — Machine learning model (Random Forest, predictions, uncertainty).*
- *[Divya Jalan] — Dashboard & visualization (Streamlit, charts, health gauges).*
- *[Siya Julka] — Report writing, presentation & engineering justification.*

APPENDIX & REFERENCES

Assumptions & Reference Material

Assumptions

Standard air properties; healthy baseline from early cycles; ISA ambient model.

Reference Material

Saravanamuttoo, “Gas Turbine Theory”; NASA C-MAPSS degradation datasets.

Tools

3D Engine Visualization Three.js + React Three Fiber	**CAD Processing** Autodesk Fusion / Blender / CAD Assistant	**Frontend** React + TypeScript	**Styling** Tailwind CSS	**Backend API** FastAPI
AI Framework PyTorch	**Dataset** NASA C-MAPSS (FD004)	**Data Processing** Pandas + NumPy	**Visualization** Plotly / Recharts	**3D Animation** React Three Drei / GSAP